

Machine Learning-Based Prediction of Diabetes Using Clinical and Demographic Risk Factors

Abstract

Diabetes is one of the most prevalent chronic diseases worldwide, making early identification of high-risk individuals essential for preventing long-term complications. Traditional screening methods rely on simple clinical thresholds and may fail to capture complex relationships among demographic and clinical risk factors. Machine learning offers a flexible framework for modeling nonlinear relationships to improve risk detection.

We evaluate Logistic Regression, Decision Tree, Random Forest, and XGBoost models using a large dataset of patient demographics and clinical biomarkers. Because missing diabetic patients is particularly costly in a screening setting, model evaluation prioritizes recall and incorporates techniques to address class imbalance. Although the Decision Tree performs competitively under default settings, Precision–Recall and ROC analysis indicates that the Random Forest has greater overall discriminative capability. After threshold tuning to achieve a target recall, the Random Forest provides the best balance between detecting diabetic cases and limiting false positives. These results demonstrate that machine learning models optimized for clinical objectives can provide more effective screening tools than traditional rule-based approaches.

1. Introduction

Diabetes is a major global health challenge, affecting hundreds of millions of individuals and frequently leading to severe complications such as cardiovascular disease and neuropathy. Early identification of high-risk individuals is therefore essential for enabling timely intervention and improving long-term patient outcomes.

Traditional screening approaches rely on clinical thresholds for biomarkers such as HbA1c and blood glucose. While these indicators are clinically meaningful, threshold-based rules often fail to capture the complex and nonlinear relationships among demographic characteristics, lifestyle behaviors, and underlying health conditions. In practice, diabetes risk emerges from multifaceted associations rather than any single measurement.

Machine learning (ML) provides a flexible framework for modeling these complex relationships. By identifying nonlinear patterns and dependencies within large datasets, ML models can learn predictive relationships directly from data rather than relying on fixed diagnostic thresholds. In

this study, we evaluate four ML models—Logistic Regression, Decision Tree, Random Forest, and XGBoost—to determine which approach best supports the high-recall requirements of a diabetes screening tool.

2. Project Objective

Our primary objective is to predict whether an individual has diabetes by leveraging a combination of demographic, behavioral, and clinical features. We formulate this task as a binary classification problem in which the outcome variable indicates the presence of the disease. Because the dataset contains a substantial imbalance between diabetic and non-diabetic cases, model evaluation emphasizes recall. In a healthcare screening context, failing to identify a diabetic patient is more harmful than incorrectly flagging a healthy individual for additional testing; accordingly, our modeling strategy prioritizes minimizing false negatives while maintaining reasonable precision and overall model stability.

More specifically, our analysis focuses on four main goals. First, we explore the relationships between demographic factors, medical history, and diabetes prevalence within the dataset. Second, we evaluate several techniques for handling class imbalance to determine which approach performs best for these data. Third, we train and compare multiple machine learning models to assess their predictive performance across standardized metrics. Then, we implement systematic classification threshold tuning to optimize the models for clinical screening and analyze model diagnostics to ensure that predictions are driven by stable and informative predictors. Finally, we examine feature importance and model diagnostics to better understand how the models make predictions.

3. Dataset Overview

The dataset used in this project contains 100,000 patient observations and 16 features describing demographic characteristics, clinical biomarkers, and medical history indicators. Initial data inspection confirmed that the dataset contains no missing values. Duplicate records were also checked and removed during preprocessing to ensure that each observation represents a unique patient entry.

The variables in the dataset can be grouped into three broad categories. The first category includes demographic information such as age, gender, location, and race. The second category includes clinical measurements such as body mass index (BMI), HbA1c level, and blood glucose level. The third category includes medical history indicators such as hypertension status, heart disease status, and smoking history. These features collectively capture multiple aspects of health that may influence diabetes risk.

The target variable indicates whether an individual has diabetes. When we examined the distribution of this variable, we found that the dataset exhibits strong class imbalance. The majority (91.5%) of observations correspond to non-diabetic individuals, while a smaller proportion (8.5%) corresponds to diabetic patients. This imbalance is important because a naïve classifier that always predicts the majority class would achieve 91.5% accuracy while failing to identify most diabetic cases.

4. Exploratory Data Analysis

We began our analysis by conducting exploratory data analysis to understand how the variables in the dataset relate to diabetes status. This step helped us identify key patterns in the data and informed later modeling decisions.

Our analysis first examined numerical variables, including age, body mass index, HbA1c level, and blood glucose level (Figure 1). We observed that diabetic patients tend to be older on average. Higher BMI values also appear to be associated with increased diabetes risk. Most notably, HbA1c levels show strong separation between diabetic and non-diabetic individuals, while blood glucose levels are significantly higher among diabetic patients. These findings are consistent with the clinical understanding of diabetes risk factors.

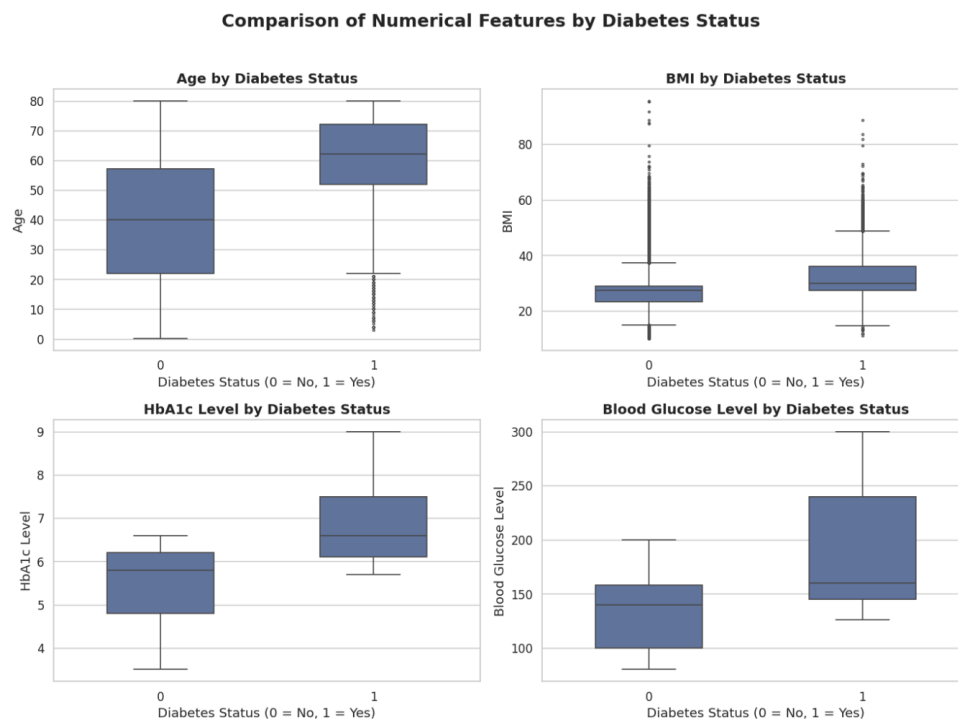


Figure 1: Distribution plots for age, BMI, HbA1c level, and blood glucose level.

We also examined correlations among variables to determine whether certain predictors showed strong relationships with diabetes status. In our analysis, some variables such as year and race appeared to show zero correlation with the target variable (Figure 2).

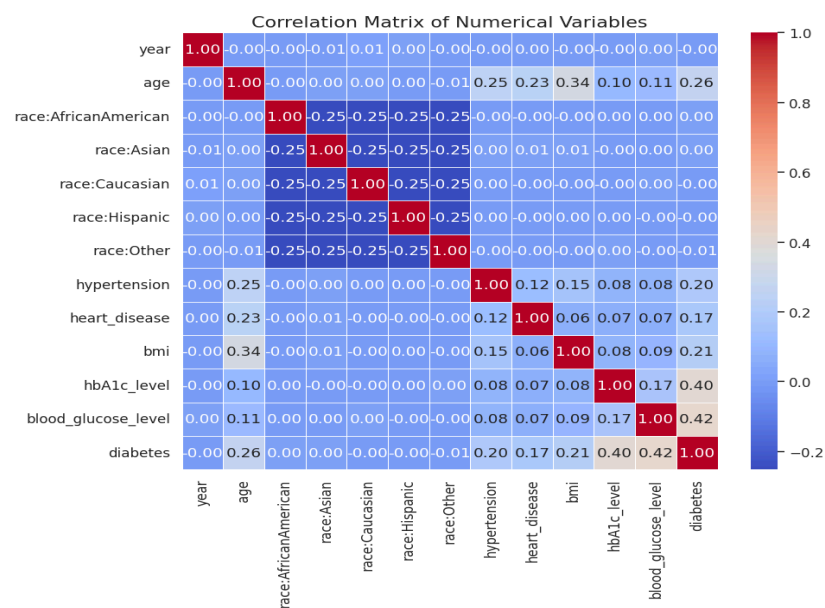


Figure 2: Correlation heatmap.

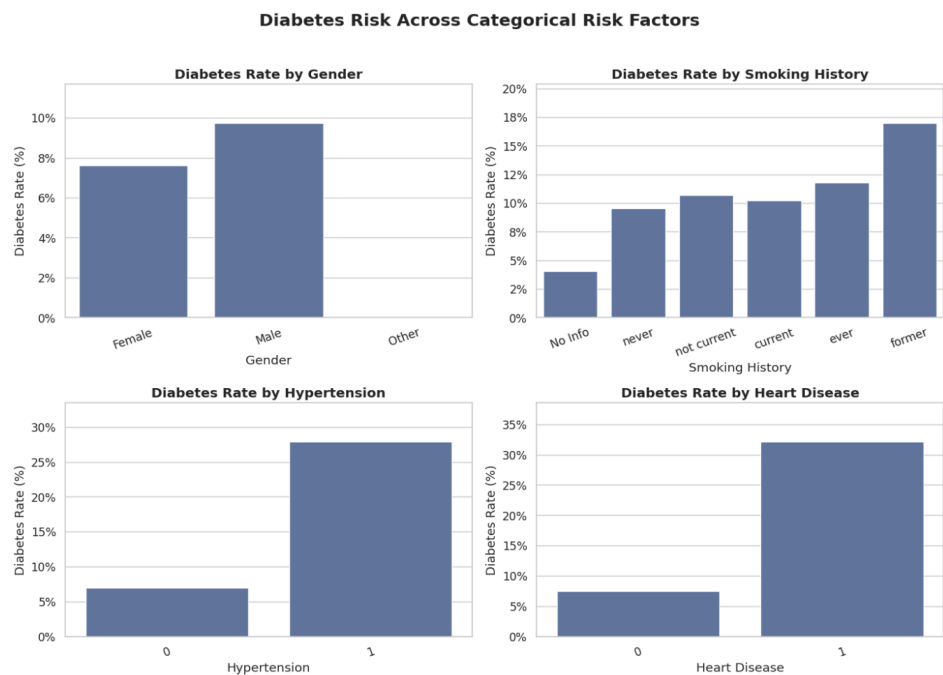


Figure 3: Diabetes prevalence by categorical risk factors.

In addition to numerical features, we analyzed several categorical risk factors. Our results show that diabetes prevalence is slightly higher among males compared to females (Figure 3). Former smokers show higher diabetes rates compared to individuals who never smoked. Hypertension is strongly associated with diabetes risk, and individuals with heart disease show a substantially higher prevalence of diabetes.

Overall, the exploratory analysis confirmed that clinical biomarkers and certain health conditions play an important role in distinguishing diabetic and non-diabetic individuals.

5. Feature Engineering and Data Preparation

Before training machine learning models, we performed several preprocessing and feature engineering steps to improve model performance. First, we removed low-impact features that did not appear to contribute meaningful predictive information in earlier analysis. This helped simplify the dataset and reduce unnecessary noise in the modeling process.

We then encoded categorical variables so that they could be used as inputs for machine learning algorithms. Because many models require numerical inputs, categorical variables such as gender and smoking history were converted into numeric representations.

We also addressed extreme values in the BMI variable by capping BMI outliers at the ninety-ninth percentile. Extremely large BMI values can disproportionately influence some models, so limiting their range helps improve stability. In addition, we grouped age into broader categories to capture life-stage patterns that may influence diabetes risk.

After completing these preprocessing steps, our final modeling dataset contained 99,982 observations and 14 numerical features.

6. Methodology

To evaluate our models, we divided the dataset into training and testing sets using an eighty–twenty stratified split. Stratification ensured that the class distribution remained consistent across both subsets despite the imbalance in the target variable.

Because the dataset contains substantially fewer diabetic cases than non-diabetic cases, handling class imbalance was an essential step in our modeling process. We therefore investigated several imbalance-handling strategies before training the final models. Specifically, we compared class weighting, random undersampling, and the Synthetic Minority Oversampling Technique (SMOTE).

Using Logistic Regression as a baseline model, all three methods produced similar predictive performance (Figure 4). We subsequently adopted class weighting for the final models; this approach penalizes minority-class errors while preserving the full dataset, thereby avoiding the data loss inherent in undersampling or the potential for synthetic noise introduced by SMOTE.

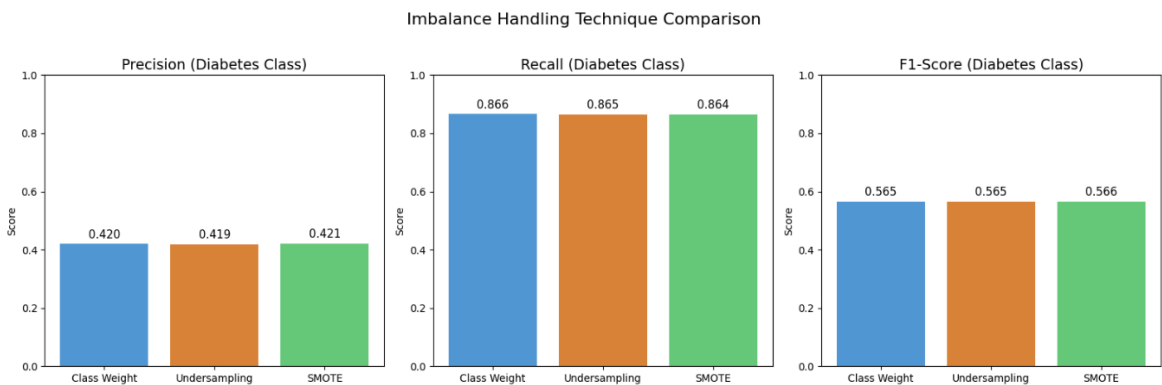


Figure 4: Comparison of class imbalance techniques.

We then trained four machine learning models with balanced class weights: Logistic Regression, Decision Tree, Random Forest, and XGBoost.

To ensure stable performance estimates, we used five-fold cross-validation during model training. Because the primary objective of this study is diabetes screening, we prioritized recall in our evaluation process. We also examined precision, F1-score, and classification thresholds to better understand the trade-offs between identifying diabetic patients and limiting false positive predictions.

7. Model Performance

We first evaluated the models under their default classification thresholds. Under these conditions, the Decision Tree achieved the highest recall, meaning it identified the largest proportion of diabetic cases (Figure 5). This improvement in recall came at the cost of lower precision and F1-score. At the same time, Random Forest and XGBoost demonstrated more stable performance across multiple evaluation metrics. However, because recall is the primary evaluation metric in this diabetes screening context, we consider the Decision Tree to perform best among the models under the default thresholds.



Figure 5: Model comparison under default thresholds.

Both Random Forest and XGBoost showed stronger discrimination when we examined their ROC and precision–recall curves, because they exhibited the largest areas under these curves (Figure 6). This indicated that their default performance was likely limited by the standard 0.5 threshold rather than a lack of predictive capacity.

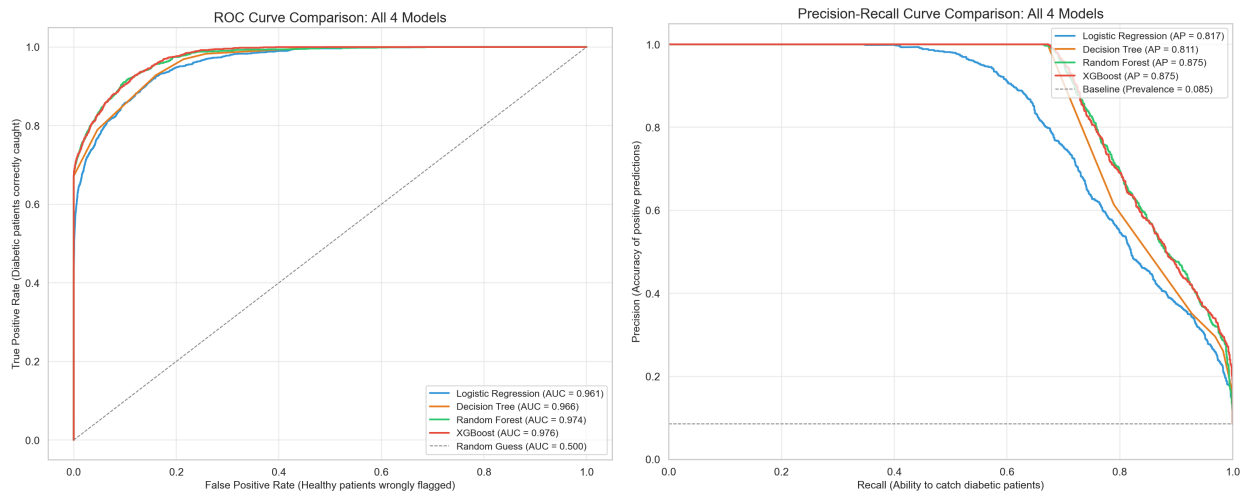


Figure 6: ROC and precision–recall curves.

8. Threshold Tuning and Final Model Selection

Because the goal of diabetes screening is to reduce missed diabetic cases, we further compared models after adjusting their classification thresholds. We tuned the thresholds so that each model achieved approximately 0.92 recall. This allowed us to compare the models at a similar recall level.

Through the systematic evaluation of performance across the full spectrum of operating points, we identified specific optimal classification thresholds for each model. The baseline Logistic Regression was optimized at a threshold of 0.38, while the XGBoost model achieved its best performance at 0.37. The tree-based architectures required slightly higher thresholds, with the Decision Tree and Random Forest models optimized at approximately 0.41 and 0.47, respectively. Adjusting these boundaries away from the default 0.5 enabled each model to better accommodate the high-sensitivity requirements of diabetes screening.

After threshold tuning, Random Forest provided the best balance across precision, F1-score, and error rates. It achieved the strongest overall performance while maintaining the target recall level (Figure 7 & 8).

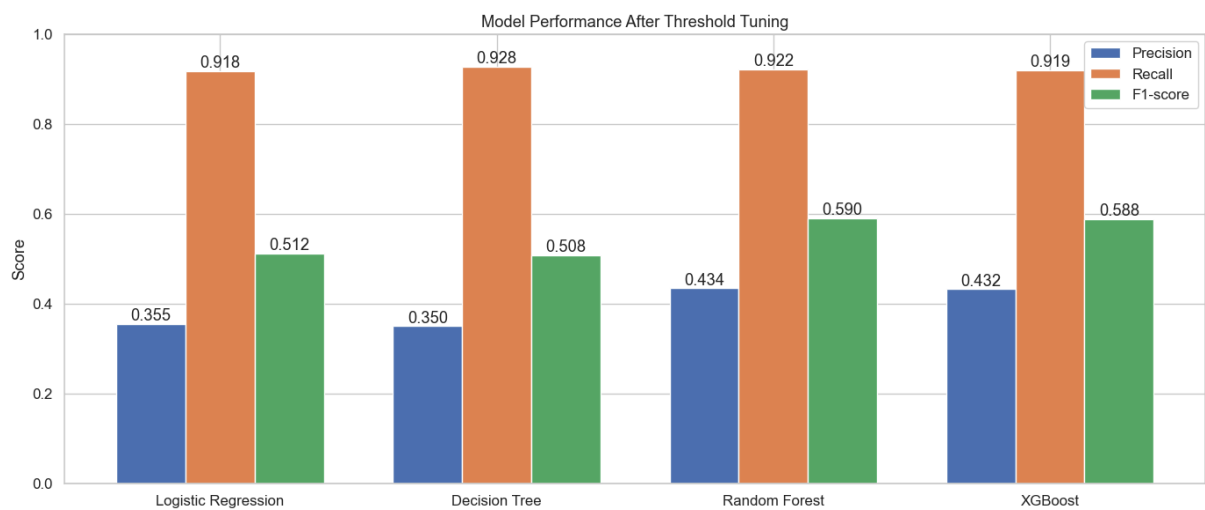


Figure 7: Model comparison under optimal thresholds.

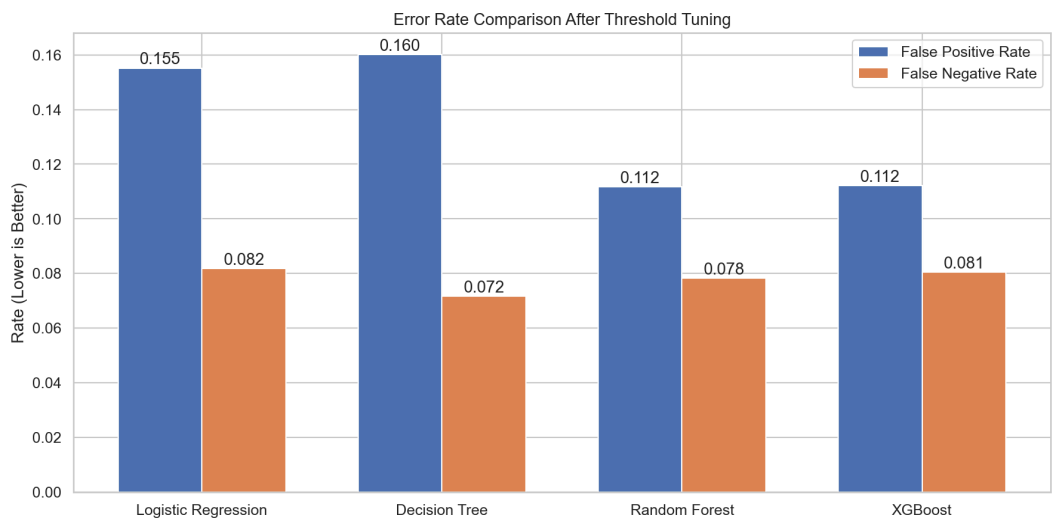


Figure 8: Final false positive and false negative comparison.

9. Model Diagnostics and Interpretation

To verify the model's internal decision-making structure and ensure it is driven by high-signal clinical variables rather than random dataset noise, we conducted a feature importance diagnostic (Figure 9). The attribution of predictive weight is heavily concentrated on established clinical biomarkers, with the HbA1c level serving as the primary driver, followed by glucose level, age, and BMI. This hierarchy is diagnostically sound because it aligns with clinical diagnostic standards, confirming that the model has captured meaningful metabolic patterns rather than relying on idiosyncratic correlations.

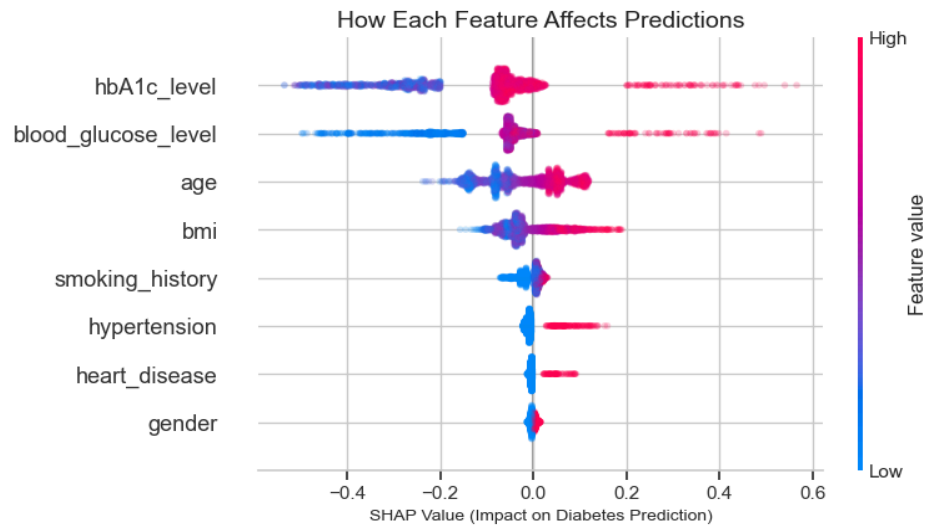


Figure 9: Feature importance plot.

To evaluate the generalizability of the final Random Forest model, a diagnostic overfitting check was performed by comparing performance across the training and testing datasets (Figure 10). The model demonstrated exceptional stability, achieving a training accuracy of 90.5% and a testing accuracy of 90.4%, representing a negligible difference of only 0.1%. More importantly, given the clinical priority of sensitivity, the recall remained robust across both subsets; the model yielded a training recall of 92.1% and a testing recall of 90.7%. The marginal difference of 1.4% in recall performance confirms that the model successfully captured generalizable biological patterns rather than overfitting to the training data, ensuring reliable performance when applied to unseen patient records.

10. Limitations and Future Work

Despite the promising performance of the optimized models, this study possesses several limitations that warrant consideration. A primary concern is the inclusion of clinical biomarkers such as HbA1c and blood glucose levels. Because these variables are the definitive markers used in clinical settings to diagnose diabetes, their presence within the feature set may lead to 'feature inflation,' potentially resulting in an overly optimistic assessment of the model's predictive power. Furthermore, from a practical standpoint, these invasive measurements require clinical testing and may not be readily available for every patient in a non-clinical screening scenario. Additionally, while multiple mitigation strategies were employed, the dataset remains fundamentally imbalanced, which continues to pose a technical challenge for minority class learning. Finally, the current analysis is constrained by a lack of behavioral data; the absence of variables related to diet, physical activity, and other lifestyle factors—which are known to significantly influence metabolic risk—precludes a truly holistic assessment of a patient's diabetic status.

Future research should address these gaps by evaluating models that exclude direct clinical markers to determine the predictive strength of non-invasive demographic data. Additionally, validating the framework on external healthcare datasets and incorporating behavioral factors would significantly enhance the model's generalizability and provide a more holistic assessment of patient risk.

11. Conclusion

In this study, we developed machine learning models to predict diabetes status by integrating demographic, lifestyle, and clinical features. Through a systematic workflow—encompassing exploratory analysis, imbalance mitigation, and comparative modeling—we identified a predictive strategy optimized for the high-sensitivity requirements of clinical screening.

Our findings reveal that while the Decision Tree achieved the highest recall under default settings, the Random Forest demonstrated superior overall discriminative capability when evaluated via Precision–Recall and ROC curves. By implementing systematic classification threshold tuning to achieve a target recall of approximately 0.92, the Random Forest provided the optimal balance between identifying diabetic cases and maintaining precision. Model diagnostics further confirmed that these predictions are primarily driven by clinical biomarkers—particularly HbA1c and blood glucose—which suggests stable predictive behavior on unseen data.

Overall, this research demonstrates that machine learning models, when optimized for specific clinical objectives, provide an effective framework for early diabetes screening and risk stratification.